
Junseok Lee

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EDUCATION

University of Toronto

Incoming, September 2026

Master of Engineering in Electrical & Computer Engineering

Toronto, Ontario, Canada

- Focus: **VLSI design, Robotics, Control Systems, Embedded Systems, and Digital control logic**

McMaster University

Completed, April 2026

Bachelor of Engineering in Electrical & Computer Engineering (Co-op)

Hamilton, Ontario, Canada

- Graduated with **Cumulative GPA: 3.2/4.0**
- **Relevant Coursework:** Logic Design, Digital Signal Processing, Circuits and Waves, Programming, Hardware Design, Web and System Development, Control Systems, SCADA Systems

SKILLS

- Programming Languages: C, C++, Python, Java, JavaScript, SystemVerilog, Verilog (FPGA), Bash, HTML/CSS
- Control & Embedded Systems: CODESYS (PLC), CAN Protocol, SCADA Systems, Arduino, Danfoss Plus+1 Guide & Service Tool, Embedded Systems, Hardware Debugging, PLC Programming, HMI Development
- Engineering Tools: AutoCAD, SolidWorks, PSpice, LTSpice, Electric Circuit Design, Quartus

WORK EXPERIENCE

Flodraulic | *Georgetown, Canada* | *CODESYS, CAN Protocol, Danfoss Plus+1, AutoCAD, Flutter*

Aug 2024 - Aug 2025

- Designed and implemented HMI system, **integrating control system and hydraulics** using CODESYS, AutoCAD, Danfoss +1 Guide/Service tool (Logic Design tool) as a Control System Engineer.
- Diagnosed and resolved **wireless system issues** (transmitters, receivers, internal logic, cable harnesses), **reducing troubleshooting time by 40%**.
- Conducted **hardware diagnostics** and implemented customer-requested program **logic updates onsite** to meet specific functional requirements.
- Developed **custom debugging solutions** tailored to client needs using **Flutter**, improving diagnostic accuracy and **enhancing user experience** for the company's app.

SROOK Pay | *South Korea* | *Google Analytics, Java, JavaScript, MySQL, HTML/CSS*

Jun 2024 - Aug 2024

- Built integrated website reports with Java/JavaScript and analyzed consumer behavior via Google Analytics, MySQL, and Google Cloud to drive data-informed decisions.

TECHNICAL PROJECTS

Fitting-room Intelligent Tracking & Transport | *Flutter, Python, STM32, FreeRTOS, AI/CV, MQTT*

Sep 2025 – Apr 2026

- Developed a **retail automation system** with an autonomous rover for fitting-room assistance and clothing delivery.
- Built customer and staff interfaces for queue tracking, room status monitoring, and service requests.
- Integrated STM32, FreeRTOS, and sensors to support safe rover operation in retail environments.

Integrated Automation Control System | *Python, Arduino, CAN BUS interface*

May 2025 – July 2025

- Developed an **Arduino-based control system** at Flodraulic, optimizing **real-time data handling and device operability via CAN bus** with dynamic protocol and baud rate adjustments.
- Engineered a real-time CAN bus system with dynamic protocol selection, improving communication efficiency.
- Achieved an **80% reduction** in device testing time through embedded diagnostics and enhanced LED/button feedback.

Pacemaker | *Python, MATLAB Simulink* [\[View Project\]](#)

Sep 2023 – Dec 2023

- Designed a Digital Circuit Model (DCM) in MATLAB Simulink to simulate the bioelectrical interface between a Python Tkinter heartbeat programming application and pulse generator, implementing multiple operational modes with real-time electrogram simulation.

RTL Image Decompression Engine | *SystemVerilog, C, Quartus*

Sep 2023 – Dec 2023

- Designed synthesizable **RTL for a full image decoder pipeline** (upsampling, YUV→RGB, IDCT) on **Altera DE2 FPGA at 50 MHz**; verified **critical path timing (4.113 ns)** via Quartus Timing Analyzer and applied **fixed-point arithmetic** ($\times 4096/\times 65536$ scaling) to replace floating-point for synthesis efficiency.
- Architected resource-constrained datapath with **4 shared multipliers**, achieving **~78% utilization** via **MAC units** and operand-multiplexed 6-tap FIR filter; designed **hierarchical FSMs** managing SRAM/UART/VGA interfaces across 796 registers.

LEADERSHIP

Korean Students Association of Canada (KSAC) | [Head of IT](#)

Jun 2022 – Present

- Managed website maintenance, security updates, and feature development to improve platform stability and user experience. (<https://ksaccanada.com/>)